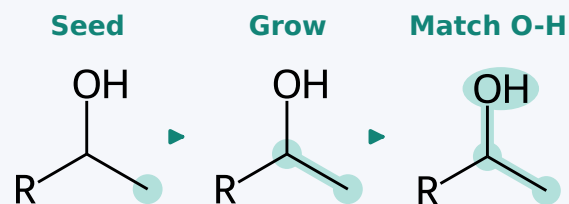
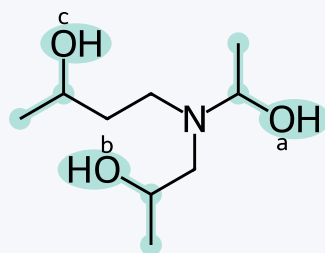


a GRAFT growth: earlier placements condition every later choice

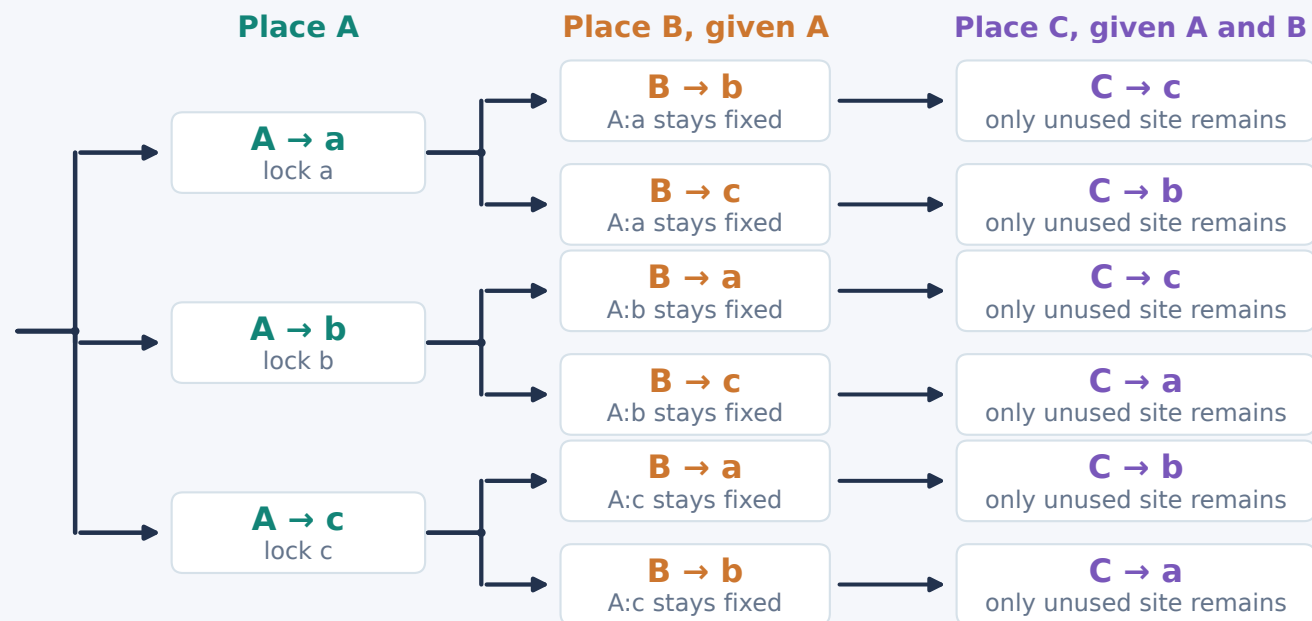
Local alcohol fragments A, B and C share this motif; R lies outside the matched atoms.



Three distinct target sites



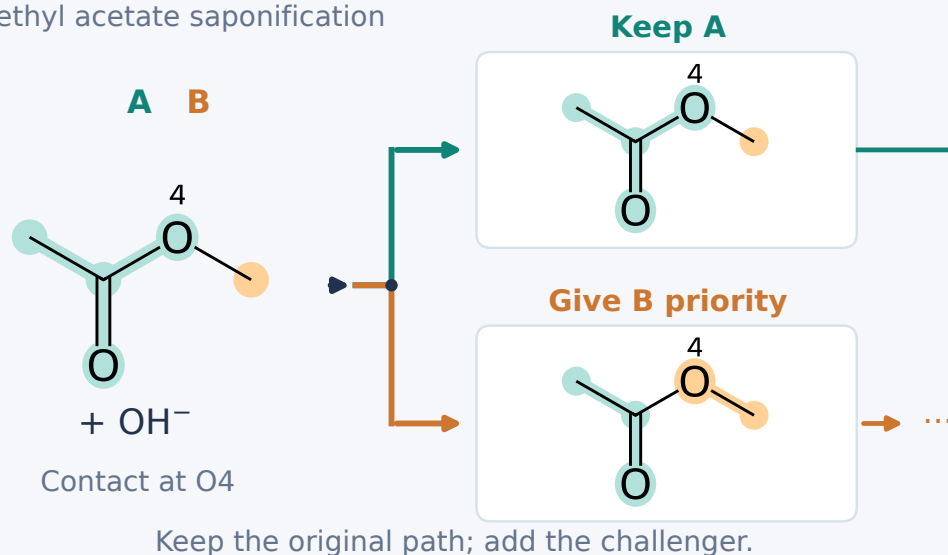
Each site preserves C-C, C-O and O-H.



Each path is a joint assignment; occupied target atoms cannot be reused.

b Add a branch when fragments compete

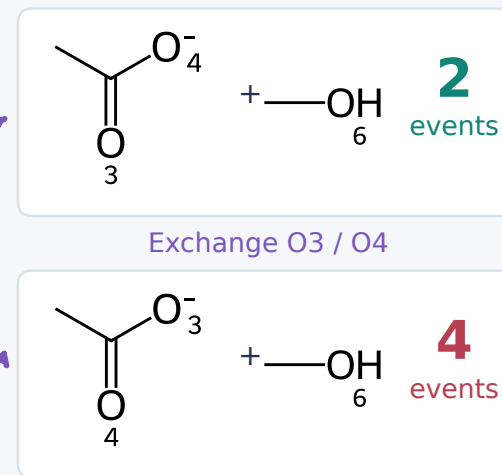
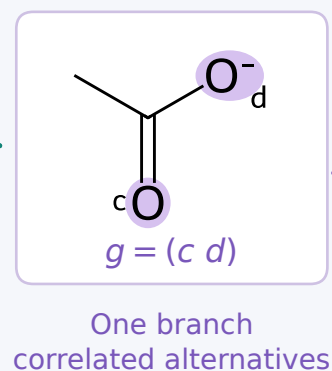
Methyl acetate saponification



c Decode each compressed family

The oxygen exchange preserves matching at $\tau_{\text{iso}} = 1$

Matching symmetry



→ Search continuation

→ Event projection

Formal bond orders · heavy-atom events at $\delta = 0.5$